

CHARCUBE · REQUIREMENTS

CharCube requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R5 relaxed to 5 h, R9 restated to 700 °C, R10 notes the safety kit; status from CCB-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	Apply Amish's 2026-09-25 decisions (CCB-DDR-001); R1 limited to the decided feedstock, R10 redefined to exclude the required safety kit, R9 probe class stated; status from CCB-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

CharCube requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs, and will be revised after co-design sessions. R1 and R10 were redefined by Amish's decisions of 2026-09-25 (CCB-DDR-001 items 2 and 4); R5 was relaxed and R9 restated by his acceptance of the TRL 3 recommendations on the same day (CCB-DDR-002 items 14 and 15). R10's target was raised to \$320 by Amish's budget top-up of 2026-09-26 (CCB-DDR-002 item 17). The status column gives the TRL 3 result from CCB-CAL-001 v0.3 (central estimate, with the range where it matters): met, not met, at risk, or not verifiable at TRL 3.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (CCB-CAL-001)
R1	Process a useful batch	10 kg or more of air-dry residue (15 % moisture or less) per batch, for the decided feedstock: bundled straw and mixed stalks and cobs, packed to about 120 kg/m³ . Loose straw is out of scope without a press (a separate project)	Retort volume and packed bulk density	Met: 12.0 kg (10.58 kg dry)
R2	Make biochar efficiently	Biochar 25 % or more of dry feed mass	Mass balance; later weighed batches	At risk: 28 % (20 to 35 %), 2.96 kg
R3	Make stable char	Retort core at 450 °C or more for 30 min or more per batch, logged; target molar H/Corg below 0.7, the European Biochar Certificate limit	Heat conduction and balance; later logged batches and lab analysis	At risk: core at 450 °C about 4.0 h after lighting (2.9 to 6.5 h), only if the annulus is kept hot
R4	Burn the gas, not vent it	All retort gas passes the annulus fire and the burner throat; visible smoke only during the first 15 min after lighting and the last 10 min of a batch; methane below 24 g per kg of char (the retort field average in Sparrevik et al. 2015)	Design review and air sizing; later emission measurement with a partner lab	Not verifiable at TRL 3: route closed, draft margin 2.4 with the baffle insert, throat holes (30) sized with 19 % spare area
R5	Fit a daily routine	Light to end of flaming in 5 h or less (relaxed from 4 h, CCB-DDR-002 item 14); char ready to unload within 16 h of lighting	Heat conduction and cooling; later logged batches	At risk: burn about 4.5 h (3.4 to 7.0 h); unload after about 8 h is met
R6	Recover heat as hot water	10 MJ or more into water per batch (60 L raised by 40 K)	Heat transfer; later measured water temperature	At risk: 11.5 MJ (9.0 to 19.1 MJ) with the spiral baffle insert and jacket blanket, about 45 K rise
R7	Keep the water circuit safe	Jacket open to the air at all times; no valve or cap that can seal it; water drawn from a tap at the base; jacket kept away from the lid by a separate support	Design review and safety checklist	Met by design
R8	Use little start-up fuel	5 kg or less of dry wood per batch	Energy balance; later weighed batches	At risk: 3.7 kg with the lid blanket (1.7 to 9.8 kg); light-up alone 1.7 kg
R9	Record each batch	Two type K channels (retort core, throat exit), class 1 probes, ±5 °C or better from 0 to 700 °C and indicative from 700 to 1,000 °C (restated, CCB-DDR-002 item 15), logged every 10 s to removable storage, 24 h on a USB power bank	Component data; design review	Met: ±3.4 °C to 700 °C with class 1 probes; energy 6 of 31 Wh
R10	Stay within the concept budget	Kiln parts \$320 or less (topped up from \$300, decided by Amish, 2026-09-26, CCB-DDR-002 item 17). The safety kit (extinguisher or water and shovel, gloves, eye protection, dust mask, about \$40) is required and listed separately, outside the	Priced BOM	Met: \$319, \$1 under; the decided baffle insert, jacket blanket and lid blanket add \$21

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (CCB-CAL-001)
		kiln budget ; this requirement and bom/bom-notes.md are where the safety kit is stated (CCB-DDR-002 item 16)		
R11	Be built and handled by a small crew	Built with hand tools, drill and angle grinder, with at most one welded part (the water jacket); no single lift above 25 kg per person	Masses from the model; design review	Met: lift unit 46.6 kg for two people (23.3 kg each); retort with char 15.2 kg
R12	Operate safely outdoors	Flue outlet 2.5 m or more above the ground; 5 m clear radius from structures and dry vegetation; no galvanized parts in the hot path	Model and BOM; safety checklist	Met: outlet 2.70 m; no galvanized lines

■ Requirements not met or at risk

- **R10 met with a \$1 margin.** The kiln parts total \$319 against the \$320 budget after adding the decided spiral baffle insert and jacket blanket (\$15, CCB-DDR-002 item 12) and lid blanket (\$6, item 13). Budget top-up to \$320 decided by Amish, 2026-09-26 (CCB-DDR-002 item 17); any price rise will put R10 at risk.
- **R5 at risk.** The central burn (about 4.5 h) meets the relaxed 5 h target, but a poorly conducting charge could take about 7 h.
- **R6 at risk.** The baffle insert and jacket blanket lift the central estimate to 11.5 MJ, but a short burn gives about 9.0 MJ, and the convection gain of the insert is an assumption until measured.
- **R8 at risk.** The lid blanket brings the central wood demand to 3.7 kg, but a long burn could need about 9.8 kg.
- **R2 and R3 at risk.** Yield and core temperature depend on feedstock and operation.
- **R4** cannot be shown without emission measurement.
- **R11** is met only with the jacket drained and two people lifting the tripod unit (23.3 kg each, close to the 25 kg limit).

■ Assumptions

- Feedstock (decided): bundled straw, maize or cotton stalks, cobs or prunings at 10 to 15 % moisture (12 % in CCB-CAL-001), packed to about 120 kg/m³; about 18.2 MJ/kg dry and 16.0 MJ/kg air-dry (higher heating value).
- Start-up fuel: air-dry wood at about 16.2 MJ/kg (lower heating value), preferably from residues or prunings.
- Char: about 20 MJ/kg and about 55 % organic carbon with about 29 % ash for mixed straw and stalks (IPCC 2019 defaults are 49 % for rice straw char and 65 % for herbaceous material).
- One batch a day for about 250 days a year.
- Ambient 10 to 40 °C, dry weather, wind below about 5 m/s.
- The kiln serves a household's or a trial plot's residue, not a whole field (see CCB-PRB-001, out of scope).